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----- Active space -----

ALVEOLAR VENTILATION AND PULMONARY CIRCULATION

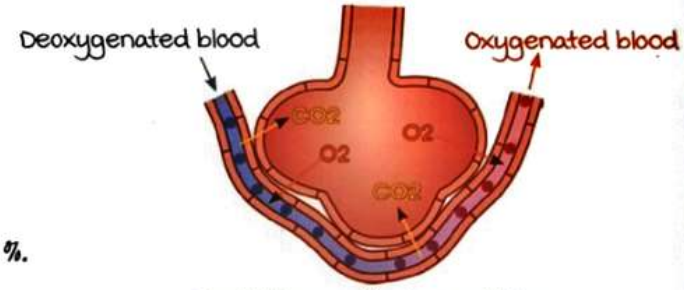
Ventilation

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USEFUL VENTILATION :

Environmental air :

- Dry.
- Pressure : 760 mmHg.
- Composition : N_a – 79 %, O_a – 21 %.



Ventilation perfusion matching
(exchange b/w O_a & CO_2)

Inspired air :

- moist.
- PO_a ↓ (D/t dilution by water vapour pressure of 47mmHg).

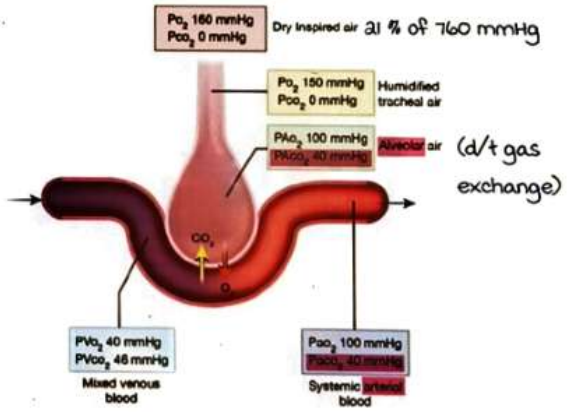
Alveolar air :

- Alveolar gas equation : $P_AO_a = PO_{i\,a} - (P_ACO_a/RER)$.

P : Partial pressure; A : Alveolar pressure; i : Inspired air.

RER : Respiratory exchange ratio :

$$\frac{\text{Volume of } CO_a \text{ produced}}{\text{Volume of } O_a \text{ consumed}} = 0.8$$



$PO_{i\,a} = FO_{i\,a}(P_b - PH_aO)$

F_i : Fractional concentration; P_b : Atmospheric pressure; PH_2O : water pressure.

– Alveolar (A) Arterial (a) Oxygen gradient = $P(A-a)O_a$.

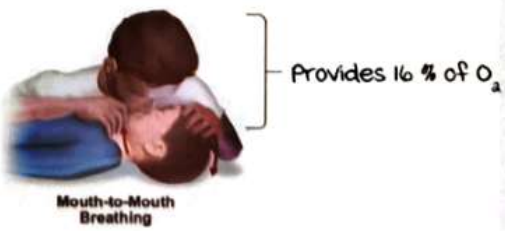
Eg : Patient with severe RDS is on ventilator, his FiO2 = 0.5, PO_a = 160 mmHg, PCO_a = 40 mmHg, barometric pressure = 747 mmHg, RER = 0.8. Calculate Alveolar O_a tension, alveolar arterial O_a gradient.

Ans : Alveolar gas equation,

$P_AO_a = 0.5 - [(747-47)-40]/0.8 = 350-50 = 300$ mmHg.

Alveolar arterial O_a gradient,

$P(A-a)O_a = 300-60 = 140$ mmHg.



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Expired air :

----- Active space -----

$P_{E}O_{2}$ = 120 mmHg.

DEAD SPACE VENTILATION :

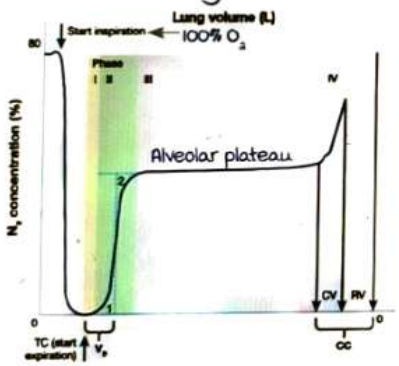
Formulas :

- Total Volume (V_T) = Dead Space (V_D) + Alveolar ventilation (V_A).
  (500 ml) (150 ml) (350 ml)
- Alveolar ventilation (V_A/min) = (V_T – V_D) × Respiratory rate (RR) [(500–150) × 12 = 4.2 L].
  - V_D/V_T = 150/500 = 30 %.
- Minute Ventilation = V_T × RR/min (500 × 12 = 6 L/min).

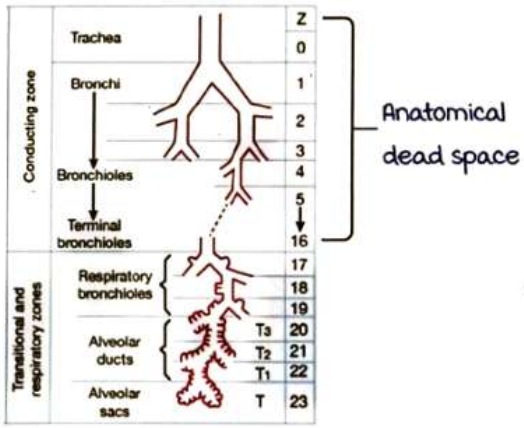
Types :

1. Anatomical dead space :

- measured by :



I – Dead space air.
II – Dead space air + alveolar air.
III – Alveolar air.
IV – Alveoli close in dependent part.
(CV – Closing volume).
RV – Residual volume.



Single breath nitrogen method/Fowler's method

- Dead space air : Volume expired b/w the beginning of the expiration & midpoint of transition phase.

2. Alveolar dead space :

- Normal : Nil.
- Pathological : Dead space occurs.

3. Total physiological dead space :

- Normal : Anatomical dead space.
- Pathological : Alveolar + Anatomical dead space.
- measured by Bohr's equation : $V_D = V_T \times \frac{P_ACO_2 - P_eCO_2}{P_ACO_2}$

Feedback

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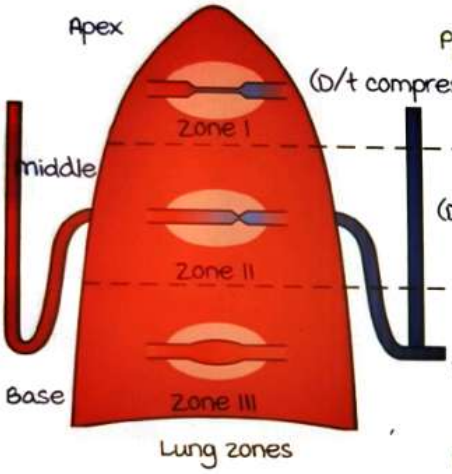


----- Active space -----

Perfusion

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Zones of perfusion :



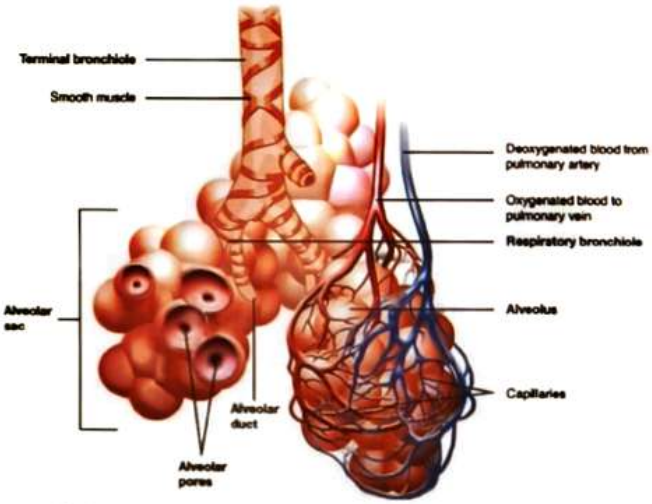
$P_A > P_a > P_v$
(D/t compression of blood vessel)

$P_a > P_A > P_v$
(D/t intermittent flow/ waterfall effect)

$P_a > P_v > P_A$
D/t continuous blood flow)

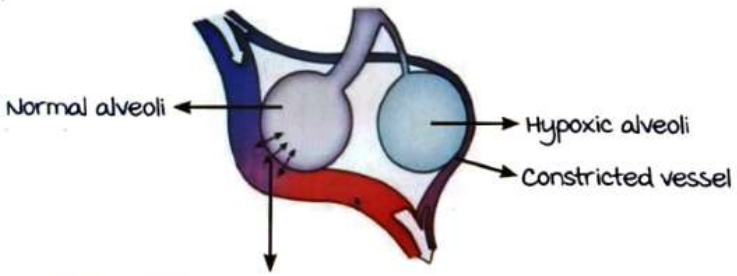
P_A : Alveolar pressure
P_a : Arterial pressure
P_v : Venous pressure

Properties of pulmonary circulation



- Pulmonary circulation = Right ventricular output = 5 L.
- Highly distensible.
- Low pressure.

Physiological response to hypoxia in pulmonary arteries :



Mechanism of vasoconstriction :

Hypoxia
↓
O_a sensitive K^+ channel closes
↓
K^+ accumulates
↓
Depolarization
↓
Ca^{a+} channel opens in vascular smooth muscle
↓
Vasoconstriction

Note : Effect of hypoxia on systemic circulation
↓
Vasodilation (eg in Kidney, heart)

Feedback